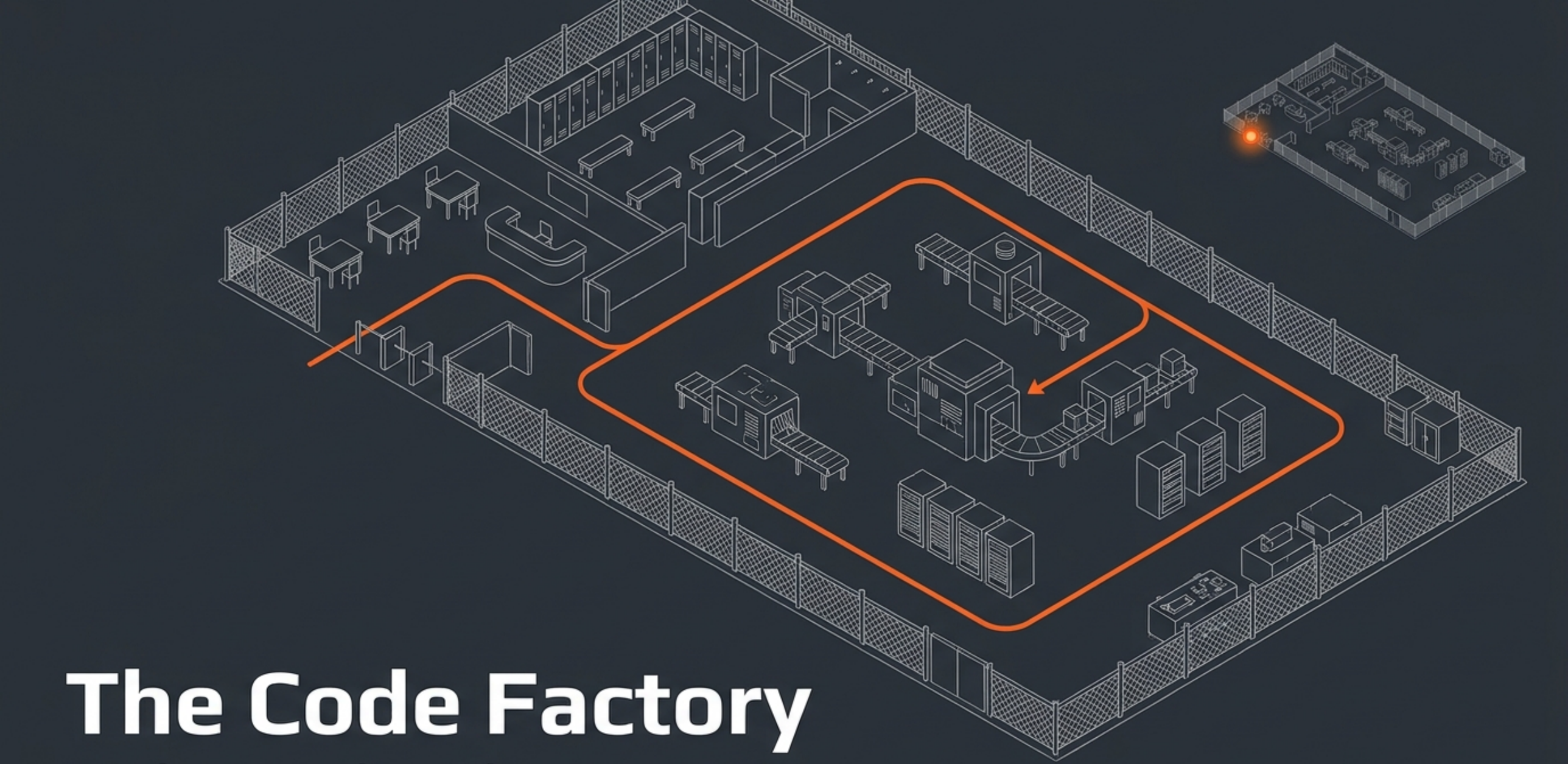
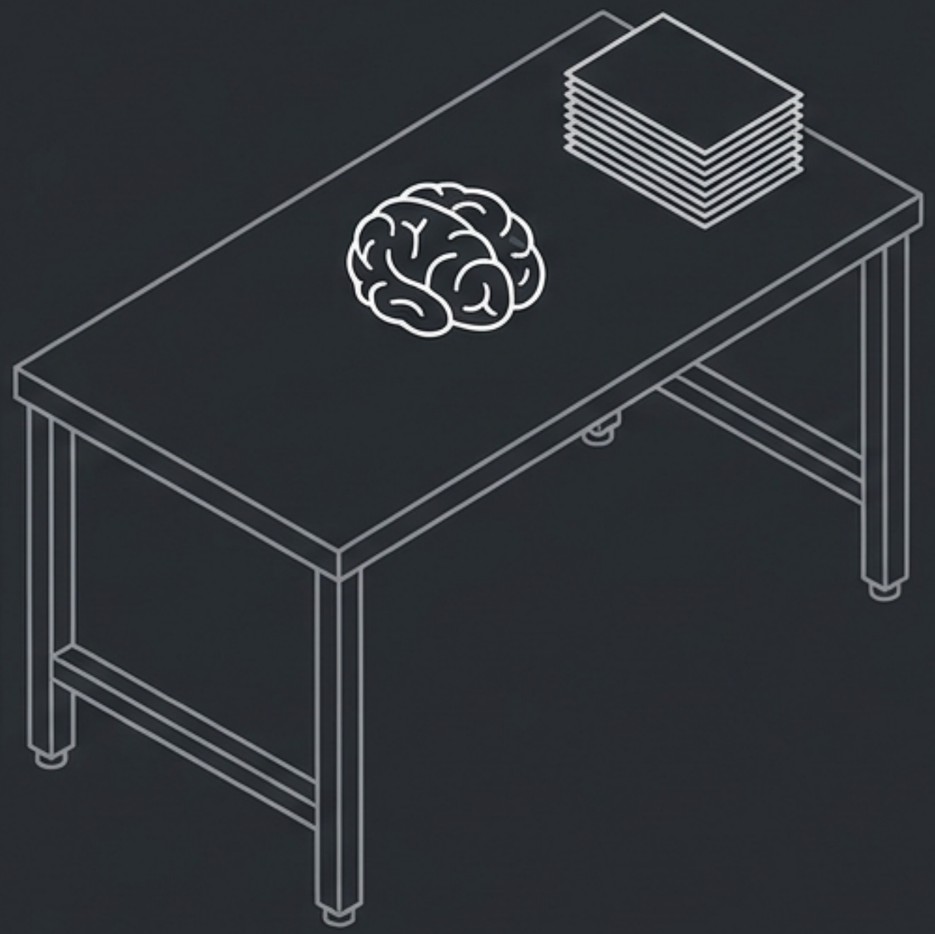


The Code Factory

Understanding the Agent Harness

Gemini Notebook





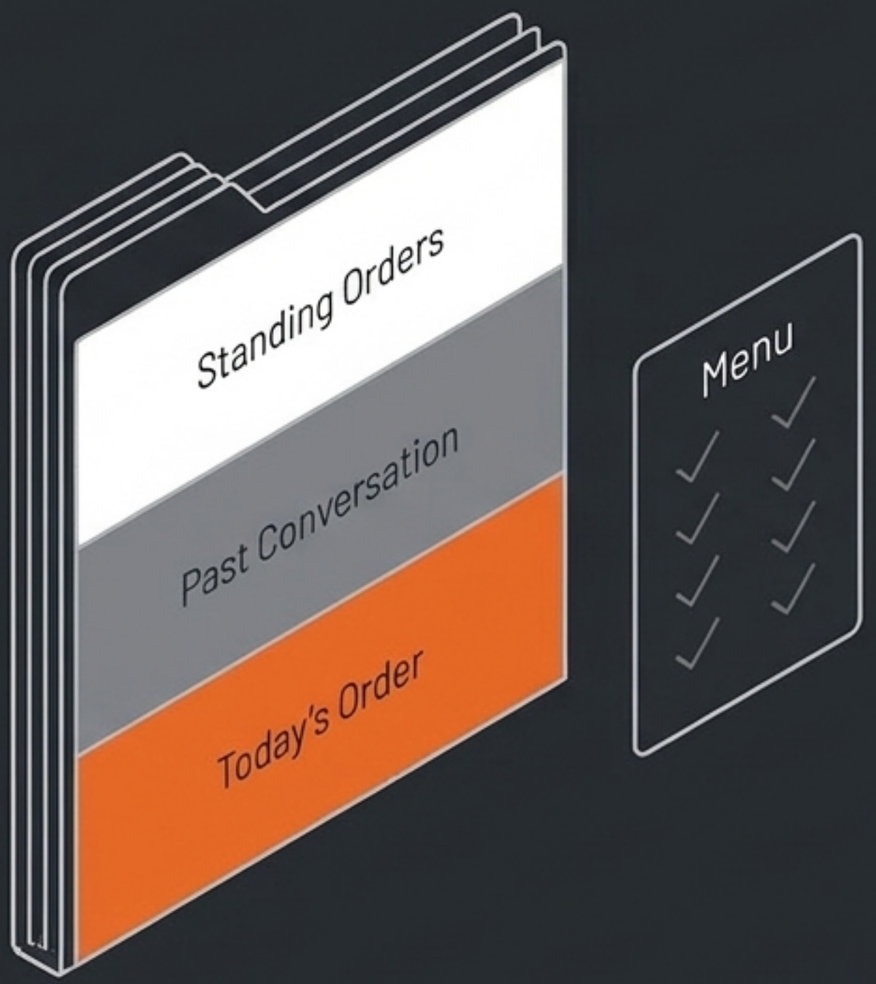
THE BRILLIANT SPECIALIST

They can solve any problem, read any document, and explain their reasoning clearly.

THE CATCH

They have no hands. They can never touch a machine. They require a surrounding system to translate judgment into action.

Gemini Notebook





The Job Folder

The Specialist's entire mind. If it isn't in this folder or on the menu, it doesn't exist.

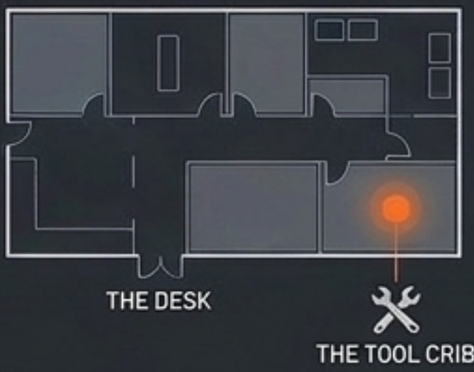
Standing Orders. How to work and what past shifts learned. (Read first, every time).

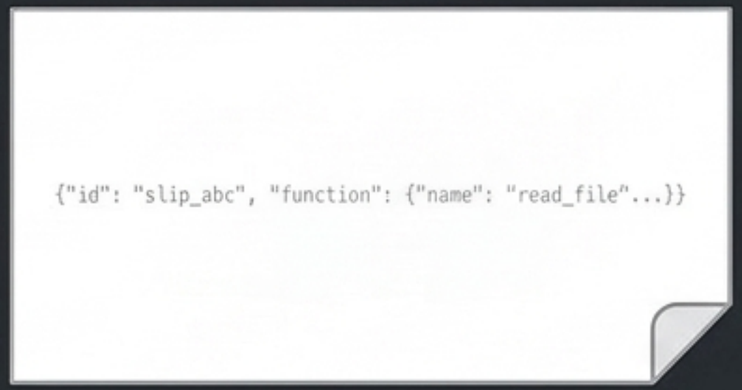
Past Conversation. The last 24 pages of earlier orders, cut cleanly.

Today's Order. The one specific task to do.

The Menu: The 8 exact types of requisition slips the specialist is allowed to write.

Gemini Notebook







Slips (Do this)

The Specialist asks for a specific action from the menu.

Receipts (Here is what happened)

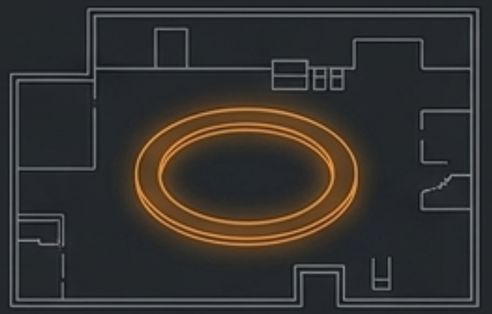
The Factory answers with the result.

The Absolute Rule: Every slip must get a matching receipt, even if a machine refuses.

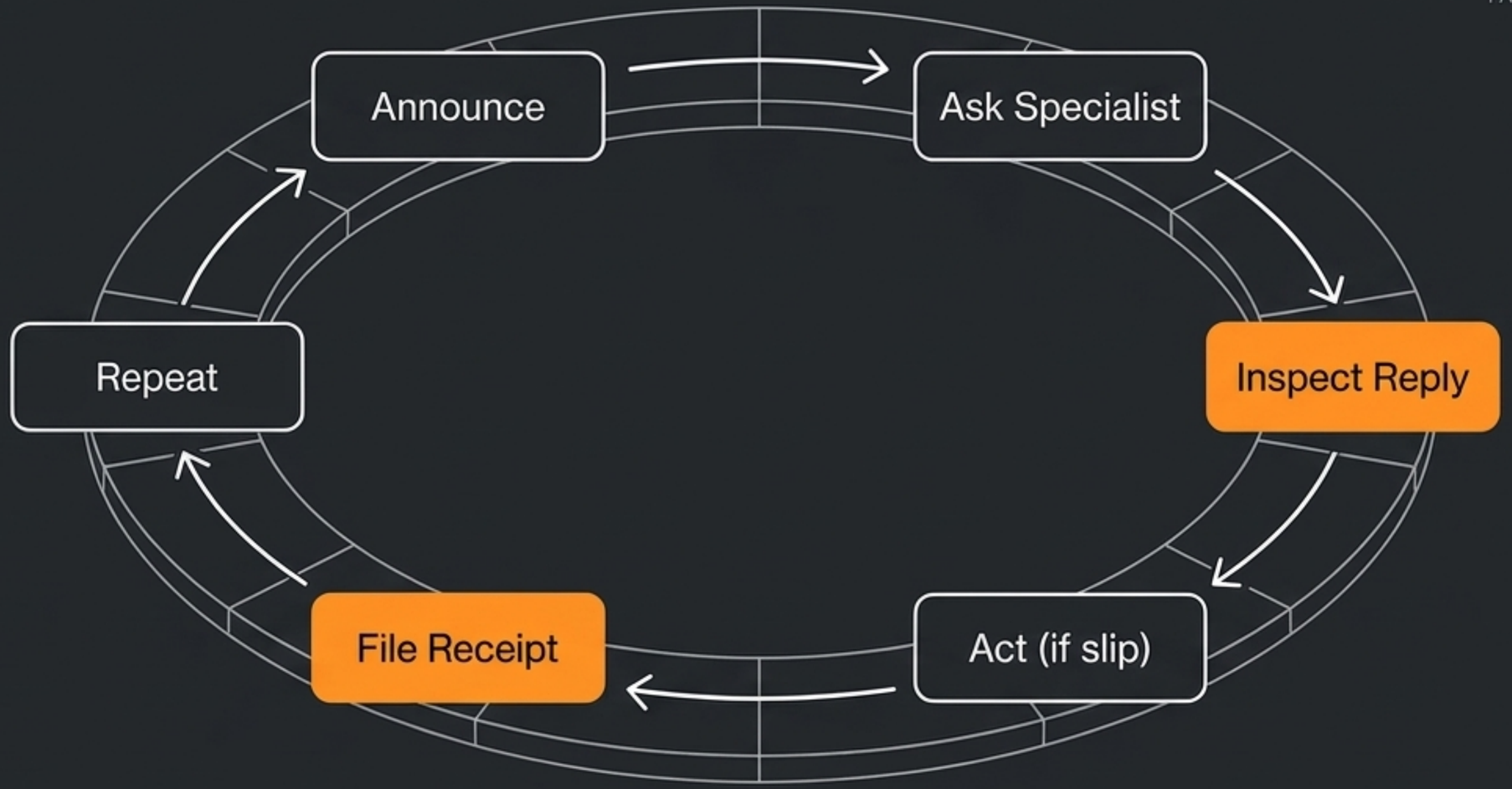
Gemini Notebook

One Order, One Loop.

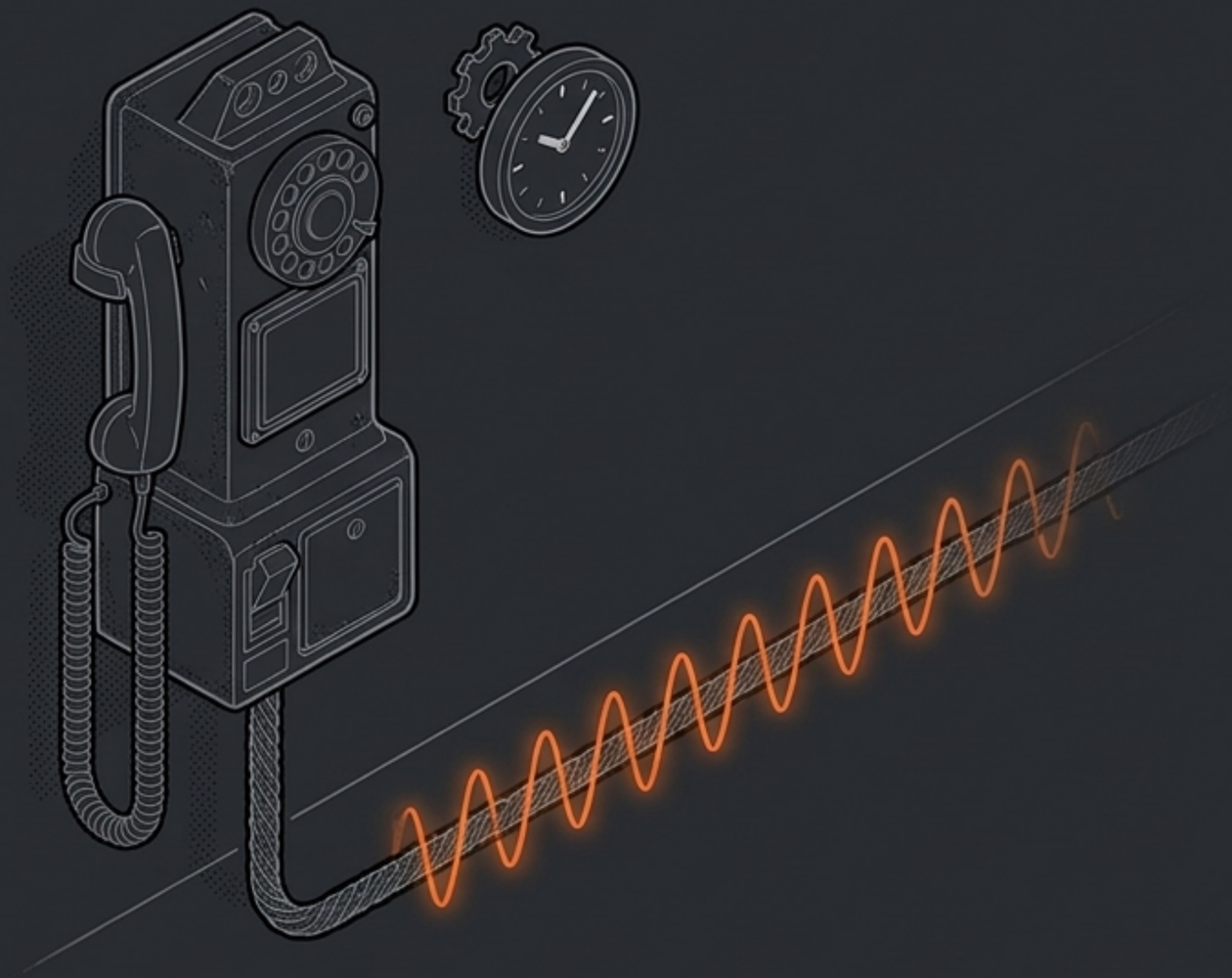
A predictable cycle to make autonomy safe, easy to watch, and hard to break.
The folder only grows during a run; nothing is thrown away mid-order.



FACTORY FLOOR



Gemini Notebook





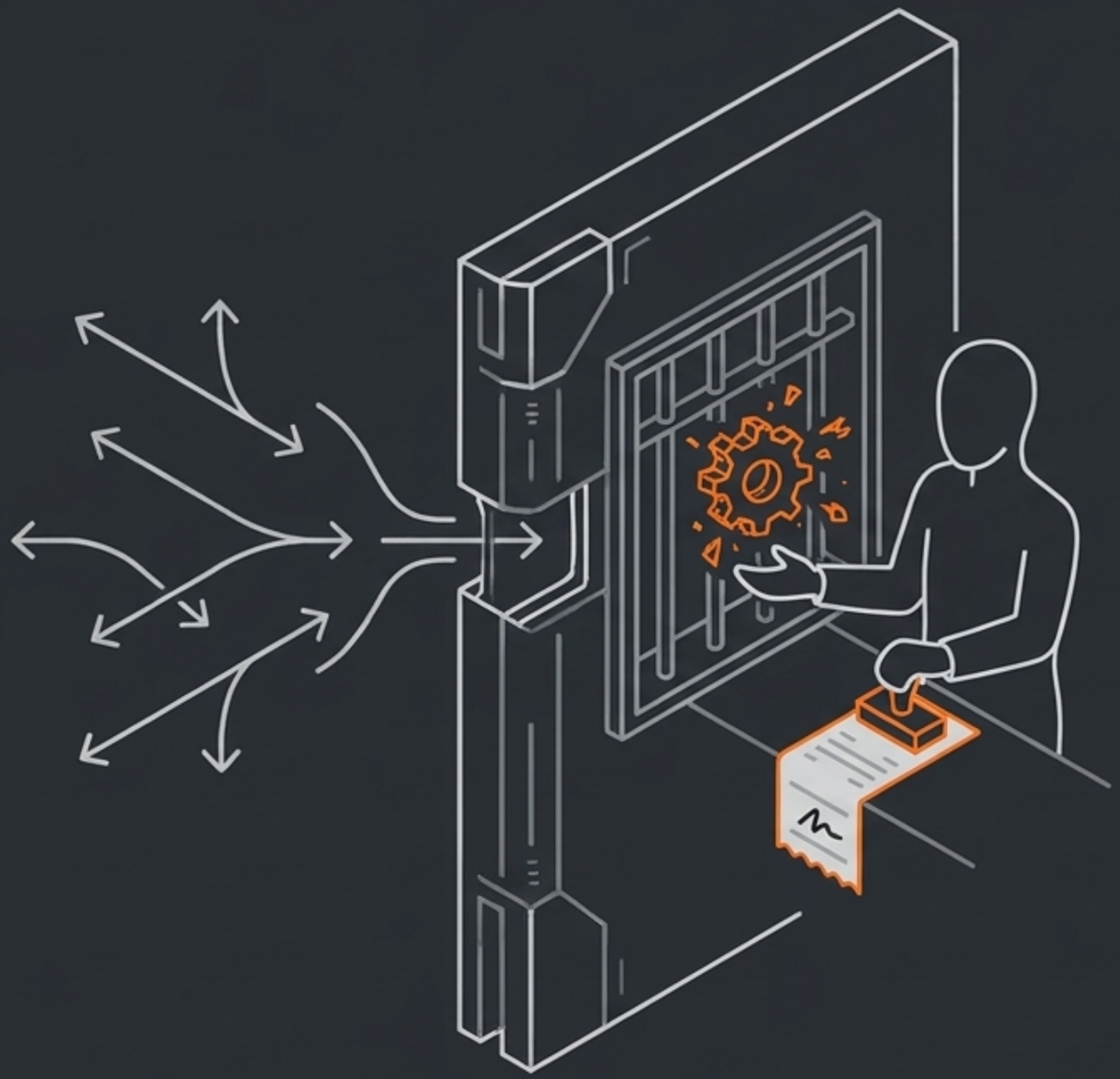
Calling Head Office

The Supervisor sends the whole folder (up to 2,048 tokens max reply) and waits for one complete answer.

The Radio Check

A slow call is not a frozen factory. Every five seconds, a signal pulses: still waiting for head office.

Gemini Notebook





The Clerk

Validates forms, operates machines, and catches everything.

Never crash, always return.

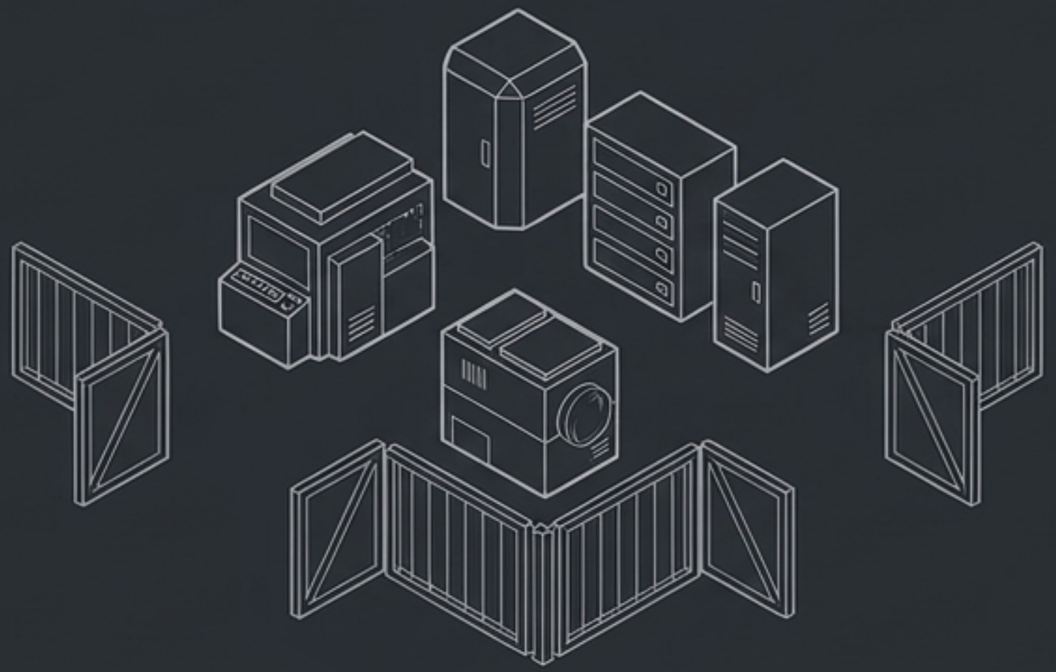
Success, bad formatting, or unexpected smoke—all become text the specialist reads on the next lap. A crash ends the shift; a receipt lets the specialist try again.

Gemini Notebook



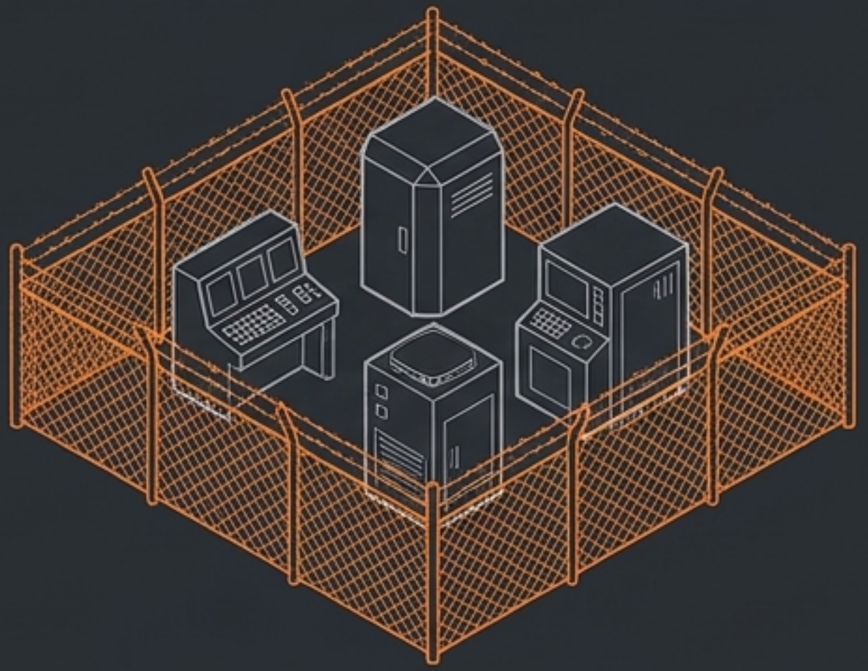
Read widely, write narrowly, report briefly.

Read-Only (list_files, read_file, recall)



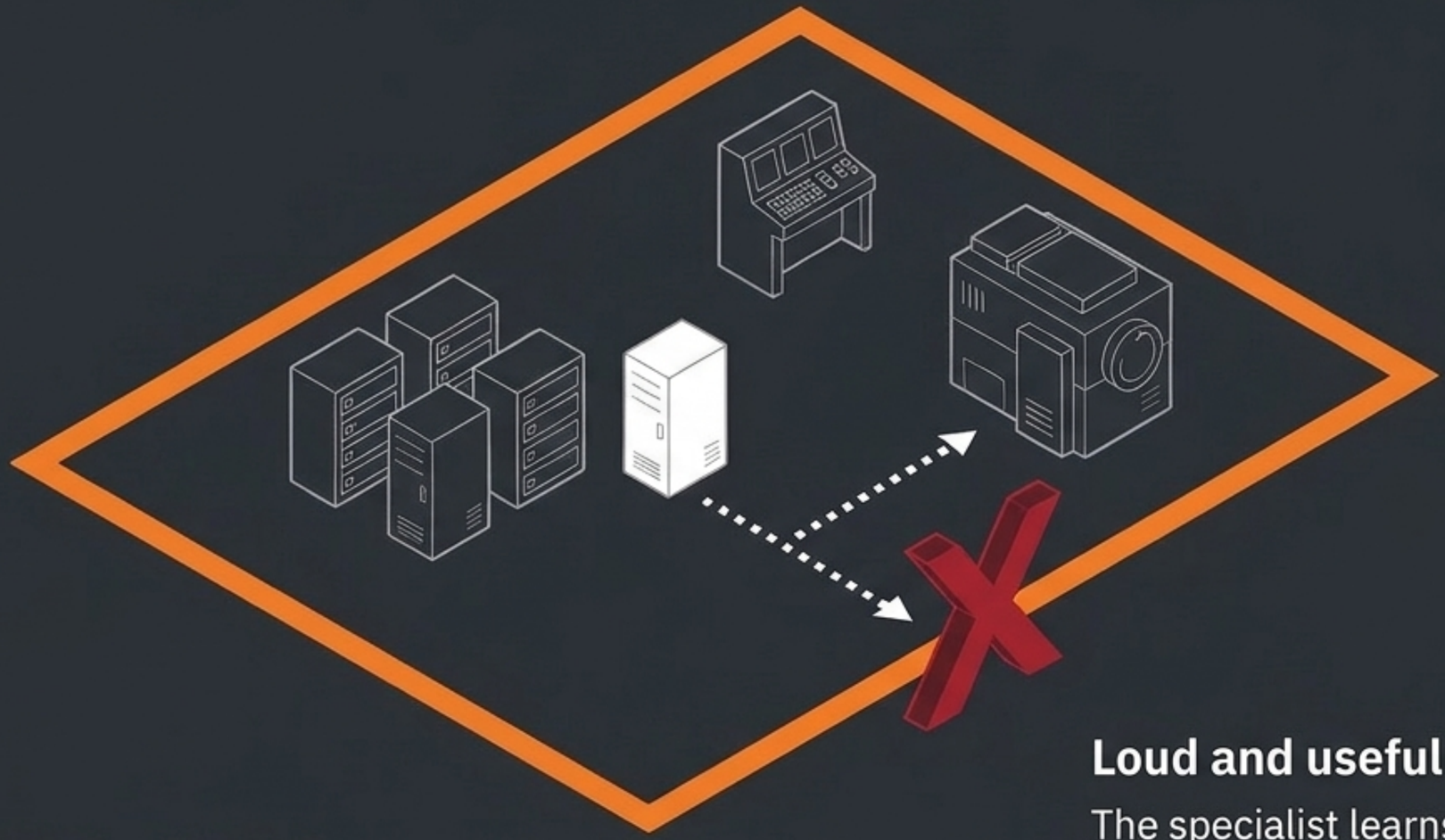
Reading is open. A specialist who cannot look at a manual is useless.

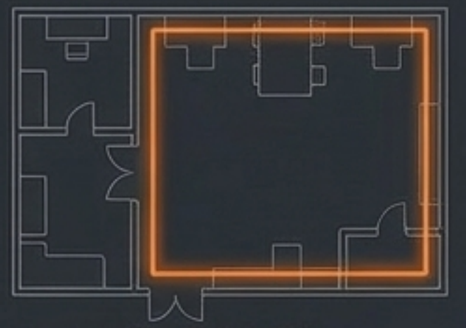
Write (write_file, edit_file, run_command)



Writing is fenced. A specialist who can put things anywhere is dangerous.

Gemini Notebook





The Painted Line.

The system refuses any file action that resolves outside the designated bay.

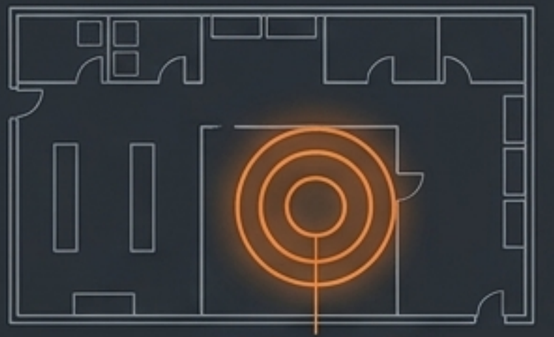
Loud and useful refusals.

The specialist learns the boundary from a clear refusal receipt, not a silent failure or stack trace.

Fences are best effort to make common mistakes loud and obvious attacks impossible.

Gemini Notebook





Defense in Depth.

Because running commands is vastly more dangerous than reading files, requiring a dedicated inspector.

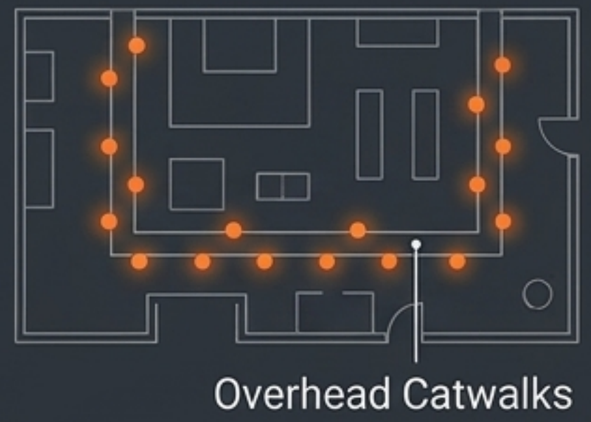
Three layers of protection ensure the system is secure before a machine even powers on. The Permit Desk checks authority, the Banned List blocks known hazards, and the Cart Inspection ensures outputs do not escape the bay.

Gemini Notebook

The Nine Gates

Checkpoints where outside staff can watch, annotate, pause, or rewrite.



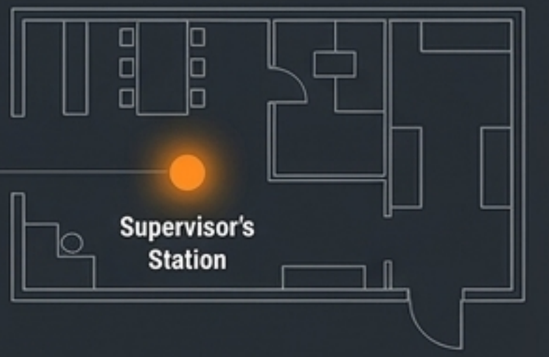


Extensibility safely.

Guards can replace a receipt or block a request, but they can never break the absolute rule: every slip still gets a receipt.

Gemini Notebook





The Question Budget

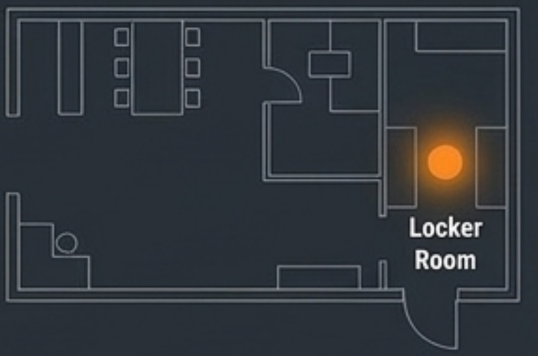
Maximum 12 calls per order.

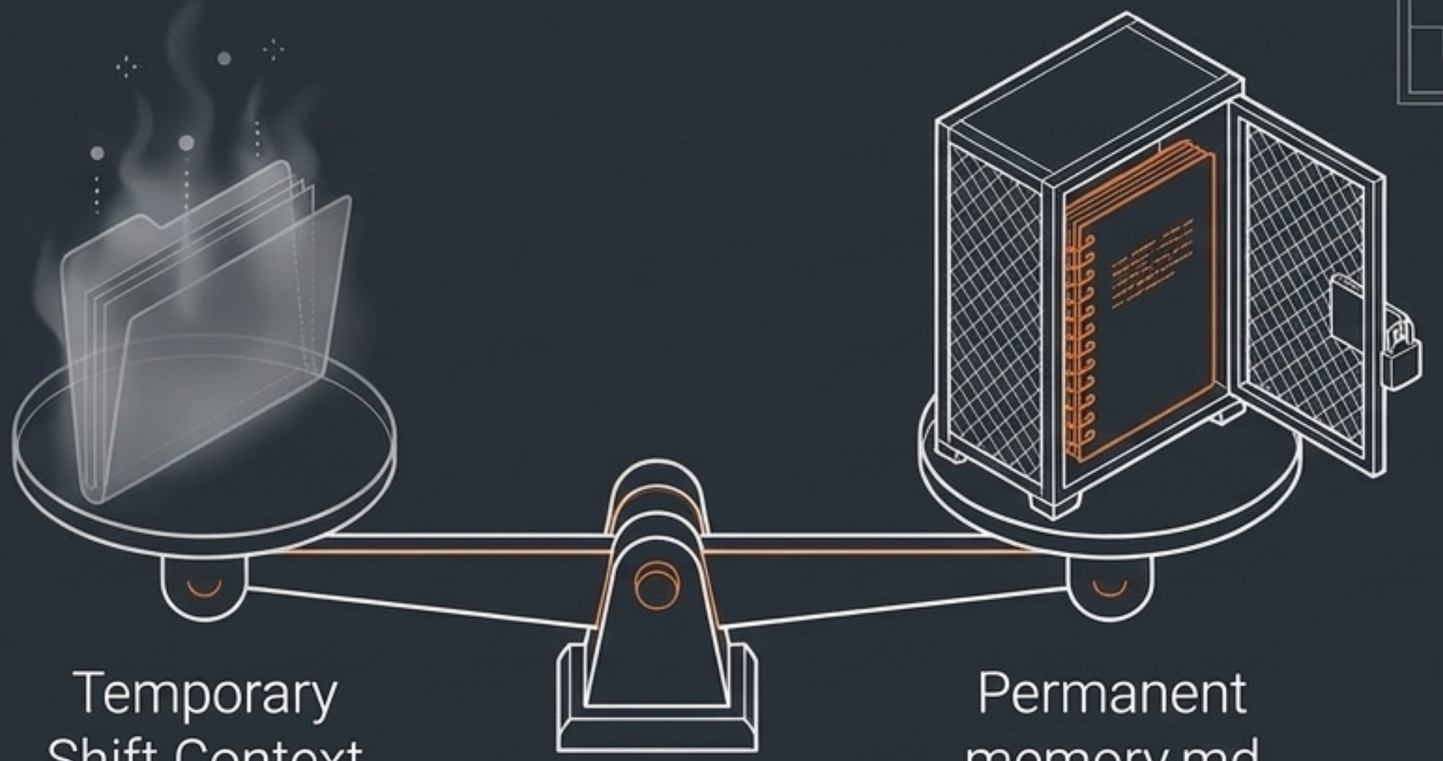
When the twelfth lap ends without a final answer, the shift stops.

Bounded autonomy is the product.

Autonomy without limits is not a feature; it is just an unbounded bill and an unrepeatable demo.

Gemini Notebook





Temporary
Shift Context

Permanent
memory.md

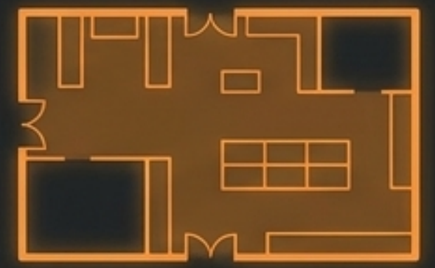
Conversation is temporary. The Notebook is permanent.

Eager digest: The newest 12 notebook lines are copied into the folder before the run starts.

Lazy recall: The specialist can spend a slip to search the full archive only when needed.

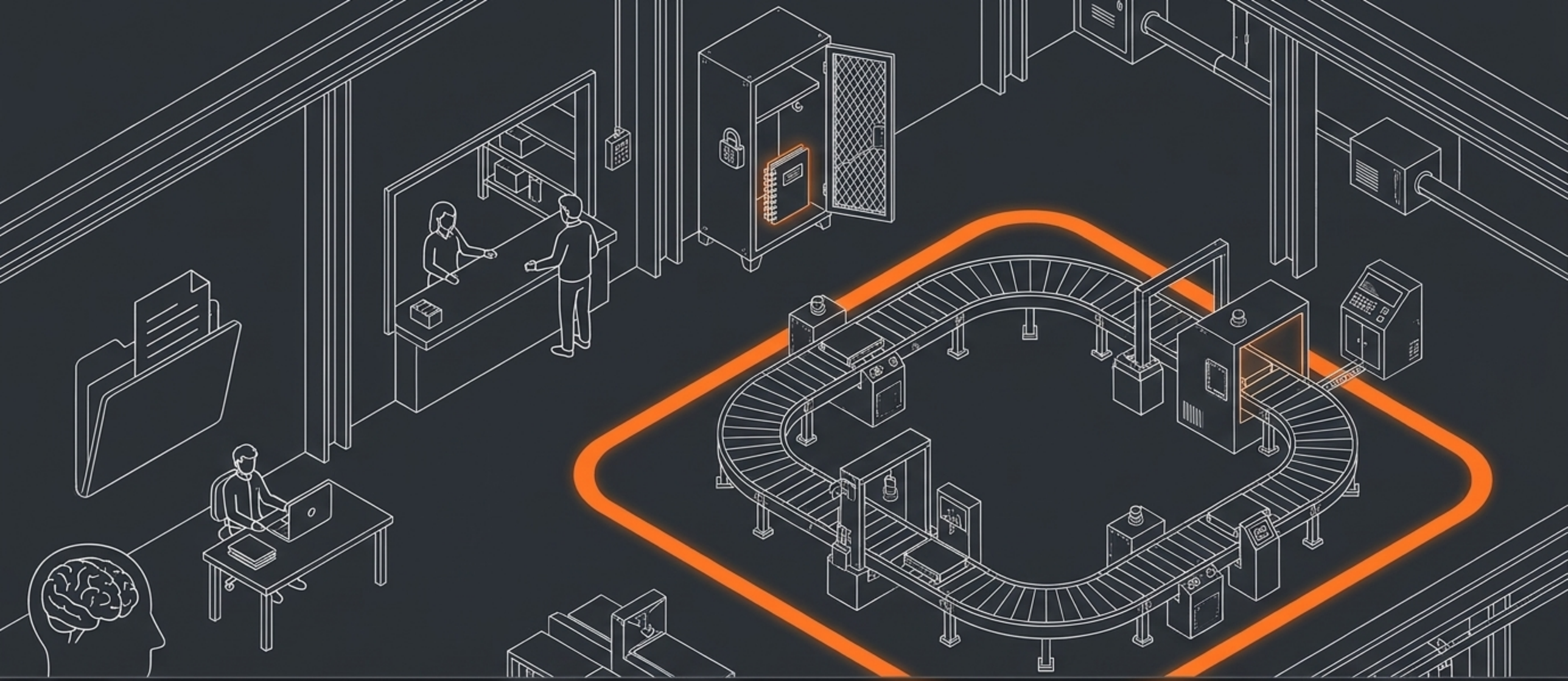
Gemini Notebook

THE BLUEPRINT TRANSLATED



FACTORY FLOOR	CODE HARNESS	ITS JOB
The Specialist	The Model	Reads the folder, writes slips or answers
The Supervisor	agent.js	Runs the lap, enforces the budget
The Job Folder	messages	The specialist's entire working state
The Painted Line	sandbox.js	Refuses writes outside the bay
The Locker Notebook	memory.md	Facts that survive the shift
The Radio Check	heartbeat.js	Proves a slow call is still alive

Gemini Notebook



A specialist asks. A clerk acts.
A line protects. A notebook remembers.

Gemini Notebook